

AIRBUS use case:

1. Predictive Maintenance

- *AI analyzes sensor data from aircraft components to predict failures before they happen, particularly engine monitoring and structural health systems*
- *Helps avoid unscheduled downtimes and improves fleet reliability.*

2. Quality Inspection with Computer Vision

- *AI-driven vision systems inspect parts for defects during manufacturing (e.g., cracks, warping, misalignment).*
- *Reduces reliance on manual inspection and speeds up production.*

3. Supply Chain Optimization

- *AI forecasts demand and optimizes logistics, parts inventory, and procurement scheduling.*
- *Mitigates disruptions and improves Just-In-Time (JIT) delivery accuracy.*

4. Knowledge Management for Documentation

- *AI parses engineering documents, manuals, and compliance requirements.*
- *Accelerates review of specs, certifications, and maintenance procedures.*
- *Helps engineers and technicians navigate large technical databases.*

5. Digital Twins for monitoring and performance testing

- *AI-powered digital twins simulate the entire lifecycle of an aircraft or subsystem.*

Supports real-time monitoring, performance testing, and predictive analytics during production and operation

6. Design Optimization (Generative Design)

- *AI algorithms generate lightweight and structurally efficient designs based on performance goals, particularly airframes, engine components, and interior parts.*
- *Reduces material usage and weight, improving fuel efficiency.*

Datasets:

1. Predictive Maintenance

Required Datasets:

- **Flight Operational Data:** Sensor logs from aircraft systems (e.g., ACARS, FOQA), encompassing flight duration, altitude, and speed.
- **Engine Health Monitoring Data:** Metrics such as vibration, exhaust gas temperature

(EGT), oil pressure/temperature, and fan speeds (N1/N2).

- **Structural Health Monitoring Data:** Readings from stress/strain sensors, crack propagation data, and assessments of fuselage/wing integrity.
- **Maintenance Logs:** Historical fault reports, work orders, and Mean Time Between Failures (MTBF) records.
- **Environmental Data:** Information pertaining to weather conditions and route-specific stressors (e.g., sand, salt, humidity).

Notes:

- This application necessitates time-series processing and anomaly detection techniques.
- Accurate data labeling to differentiate between “failure” and “non-failure” states is paramount.

2. Quality Inspection with Computer Vision Required Datasets:

- **High-Resolution Images/Videos:** Visual data of parts captured during or after manufacturing, including various perspectives.
- **Defect Annotations:** Labeled data identifying defect types, such as cracks, warping, corrosion, Foreign Object Debris (FOD), and misalignments.
- **CAD Models and Reference Images:** Ground truth data for comparative analysis.
- **Lighting/Environment Variants:** Image data acquired under diverse lighting conditions, orientations, and cleanliness levels.

Notes:

- Supervised learning methodologies, utilizing labeled image datasets, are employed.
- The incorporation of 3D data or depth maps can enhance the analysis of intricate components.

3. Supply Chain Optimization Required Datasets:

- **Historical Procurement Data:** Purchase orders, lead times, and vendor reliability metrics.
- **Inventory Data:** Stock levels, warehouse throughput, and part lifecycles.
- **Logistics Data:** Shipment tracking information, delivery times, and records of disruptions.
- **Demand Forecasts and Production Schedules:** Data reflecting seasonality and historical demand for aircraft parts.
- **External Events:** Information on market trends, geopolitical disruptions, industrial strikes, and material shortages.

Notes:

- Integration with Enterprise Resource Planning (ERP) systems such as SAP and Oracle is commonly observed.

- Reinforcement learning or time-series forecasting techniques are utilized for optimizing inventory management.

4. Knowledge Management for Documentation Required Datasets:

- **Engineering Documents:** Technical manuals, design documents, parts catalogs, and Supplemental Type Certificates (STC).
- **Maintenance Manuals:** Aircraft Maintenance Manual (AMM), Illustrated Parts Catalog (IPC), and Component Maintenance Manual (CMM).
- **Regulatory Documents:** Federal Aviation Administration (FAA)/European Union Aviation Safety Agency (EASA) compliance files and airworthiness directives.
- **Past Queries & Navigation Logs:** Data on technician search behavior and FAQ-style knowledge graphs.

Notes:

- Natural Language Processing (NLP) techniques, including document classification, summarization, and semantic search, are applied.
- The development of ontologies for aviation-specific terminology is beneficial.

5. Digital Twins for Monitoring and Performance Testing Required Datasets:

- **CAD/CAE/CFD Models:** Comprehensive 3D models and simulations of systems and subsystems.
- **Sensor Data Streams:** Real-time data from aircraft for live synchronization.
- **Material Properties:** Thermal, electrical, and mechanical characteristics of materials.
- **Maintenance & Performance History:** Correlation of real-world behavior with simulated behavior.

Notes:

- This application requires high-fidelity simulations and continuous synchronization with operational data.
- Integration with Internet of Things (IoT) frameworks and potentially edge computing is necessary.

6. Design Optimization (Generative Design) Required Datasets:

- **Design Parameters:** Constraints, load conditions, and part dimensions.
- **Material Data:** Properties for composites, alloys, and other relevant materials.
- **Performance Metrics:** Data pertaining to drag, stress points, and fatigue life.
- **Historical Designs and Results:** Previous airframe/engine/component designs and their associated test data.
- **Regulatory Constraints:** Certification requirements and design rules.

Notes:

- Reinforcement learning or genetic algorithms are employed.
- This process is often integrated with Computer-Aided Design (CAD) tools such as Siemens NX and Dassault CATIA.

Final Considerations Cross-Cuttingting Dataset Needs:

- **Metadata Repositories:** Centralized data catalogs for ensuring traceability and version control.
- **Access Control Data:** Particularly crucial for Intellectual Property (IP)-sensitive datasets within the defense and aerospace sectors.
- **Synthetic Datasets:** When real data is scarce (e.g., failure data), simulation-generated or Generative Adversarial Network (GAN)-augmented datasets can provide supplementary information.

Data Governance:

- Adherence to General Data Protection Regulation (GDPR), International Traffic in Arms Regulations (ITAR), and internal Airbus compliance standards is essential.
- Consideration of cloud-hybrid models for sensitive IP is advisable.